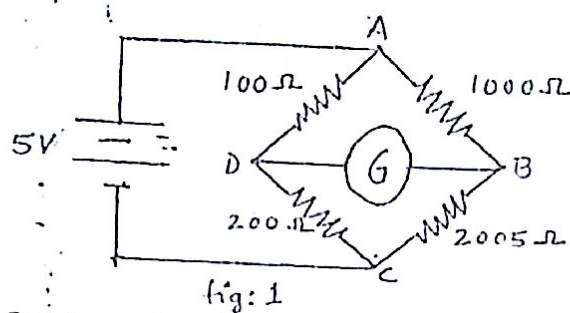


Tutorial #2

Submission Date: ~~_____~~

- 1) The schematic diagram of a Wheatstone bridge with values of the bridge elements as shown in the fig:1. The battery voltage is 5V and its internal resistance negligible. The galvanometer has a current sensitivity of $10\text{mm}/\mu\text{A}$ and an internal resistance of $100\ \Omega$. Calculate the deflection of the galvanometer caused by the $5\ \Omega$ unbalance in arm BC.

(33.2mm)



- 2) The galvanometer of Q.1 is replaced by one with an internal resistance of $500\ \Omega$ and a current sensitivity of $1\text{mm}/\mu\text{A}$. Assuming that a deflection of 1mm can be observed on the galvanometer scale, determine if this new galvanometer is capable of detecting the $5\ \Omega$ unbalance in arm BC of fig:1.

(2.24mm)

- 3) The impedances of the basic ac bridge of fig:2 are given as follows:

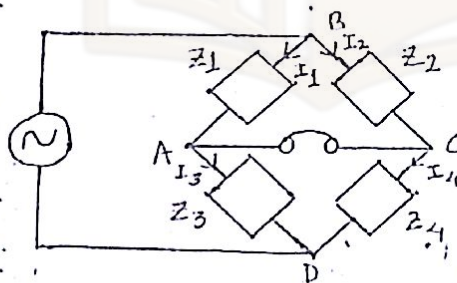
$$Z_1 = 100\ \Omega \angle 80^\circ \text{ (inductive impedance)}$$

$$Z_2 = 250\ \Omega$$

$$Z_3 = 400\ \Omega \angle 30^\circ \text{ (inductive impedance)}$$

$$Z_4 = \text{unknown}$$

Determine the constants of the unknown arm.

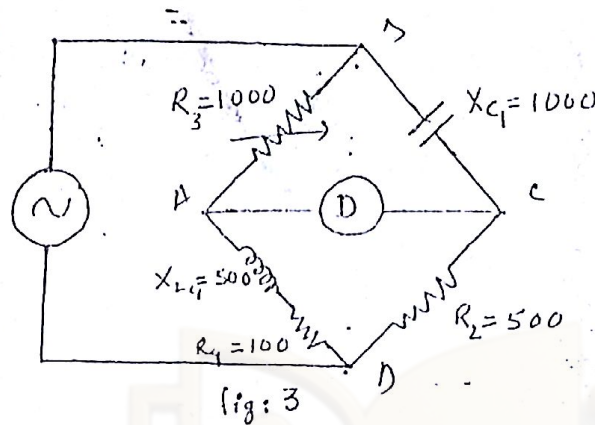


- 4) The ac bridge of fig.2 is in balance with the following constants: arm AB, $R = 450\ \Omega$; arm BC, $R = 300\ \Omega$ in series with $C = 0.265\ \mu\text{F}$; arm CD, unknown; arm DA, $R = 200\ \Omega$ in series with $L = 15.9\ \text{mH}$. The oscillator frequency is $1\ \text{kHz}$. Find the constants of arm CD.

(23.9mH)

- 5) The ac bridge of fig.2 is in balance with the following constants: arm AB, $R = 2,000\ \Omega$ in parallel with $C = 0.047\ \mu\text{F}$; arm BC, $R = 1,000\ \Omega$ in series with $C = 0.47\ \mu\text{F}$; arm CD, unknown; arm DA, $C = 0.5\ \mu\text{F}$. The oscillator frequency is $1\ \text{kHz}$. Find the constants of arm CD.

- 6) A bridge is balanced at 1 kHz and has the following constants: arm AB, $0.2 \mu\text{F}$ pure capacitance; BC, 500Ω pure resistance; CD, unknown; DA, $R = 300\Omega$ in parallel with $C = 0.1 \mu\text{F}$. Find the R and C or L constants of arm CD, considered as a series circuit. (34.3 Ω , 29mH)
- 7) A 1,000Hz bridge has the following constants: arm AB, $R = 1,000\Omega$ in parallel $C = 0.5 \mu\text{F}$; BC, $R = 1,000\Omega$ in series with $C = 0.5 \mu\text{F}$; CD, $L = 30 \text{ mH}$ in series with $R = 200\Omega$. Find the constants of arm DA to balance the bridge. Express the result as a pure R in series with a pure C or L and also as a pure R in parallel with a pure C or L.
- 8) Consider the circuit of fig.3 and determine whether or not the bridge is in complete balance. If not, show two ways in which it can be made to balance and specify numerical values for any additional components. Assume that bridge arm 4 is the unknown that cannot be modified.



- 9) An ac bridge has in arm AB a pure capacitance of $0.2 \mu\text{F}$; in arm BC, a pure resistance of 500Ω ; in arm CD, a series combination of $R = 50\Omega$ and $L = 0.1 \text{ H}$. Arm DA consists of a capacitor $C = 0.4 \mu\text{F}$ in series with a variable resistor R_s , $\omega = 5,000 \text{ rad/s}$.
- Find the value of R_s to obtain bridge balance
 - Can complete balance be attained by the adjustment of R_s ? If not, specify the position and value of an adjustable resistance to complete balance. (1000 Ω , 250 Ω)
- 10) An ac bridge has the following constants; arm AB, $R = 1,000\Omega$ in parallel with $C = 0.158 \mu\text{F}$; BC, $R = 1,000\Omega$; CD, $R = 500\Omega$; DA, $C = 0.636 \mu\text{F}$ in series with an unknown resistance. Find the frequency for which this bridge is in balance and determine the value of the resistance in arm DA to produce this balance.

Q. What ~~is~~ do you mean by loading effect? Explain loading effect in terms of series and shunt (parallel) connected instruments.

$$R \parallel \frac{1}{j\omega C} = \frac{R \times \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{R}{1 + j\omega RC}$$